

# Comparing two dead-reckoning methods on smart-ring brushing data: the in-house heuristic versus a faithful port of AEOLUS [1]

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## Abstract

The **ringbrush-coverage** project visualises toothbrushing coverage from smart-ring inertial logs. A small dead-reckoning (DR) step inside the analysis pipeline nudges the brush cursor in response to short-term motion. The production pipeline ships with an in-house *heuristic* integrator. In this note we add a faithful port of the AEOLUS dead-reckoning pipeline described in Sections 3.3–3.6 of Radeta et al. [1], and compare the two methods on four labelled sessions. The heuristic captures repetitive oscillation cleanly but yields non-metric output. AEOLUS produces metric trajectories but collapses to near-1D drift on short, repetitive brushing clips because the brush almost never crosses the paper’s zero-velocity-update (ZVU) threshold. On full-session data, both methods recover a similarly-structured path. We discuss the convention assumptions and tuning choices that shape this comparison.

## 1 Background

The **ringbrush-coverage** pipeline parses smart-ring CSV logs of the form  $\langle t_{ms}, \text{roll}, \text{pitch}, \text{yaw}, a_x, a_y, a_z \rangle$ , classifies each analysis window into one of six mouth-region labels, and accumulates per-zone coverage seconds. A short per-window dead-reckoning pass computes a tiny displacement  $(\Delta x, \Delta y)$  that perturbs the classifier’s anchor cursor so the rendered brush tip jitters in the direction the user is actually moving. The DR result is small by design: it is a visual aid, not a metric trajectory.

Radeta et al. [1] systematically evaluate fifteen dead-reckoning algorithms in underwater and ground environments using their AEOLUS apparatus (an MPU-9250 IMU on a LoPy4). Their pipeline is built around the sensor-fusion-then-double-integration recipe: orientation estimation (Madgwick by default), Earth-frame gravity removal, double integration with a ZVU-style drift reduction step, and a final heading-projected position update.

## 2 The two methods

We use the following notation. At each time step  $i$ , the IMU reports Euler angles  $(\phi_i, \theta_i, \psi_i) = (\text{roll}_i, \text{pitch}_i, \text{yaw}_i)$  in degrees and a body-frame acceleration vector  $\mathbf{a}_i = (a_{x,i}, a_{y,i}, a_{z,i})$  in  $\text{m/s}^2$ . The time step is  $\Delta t_i = (t_i - t_{i-1})/1000$  s.

### 2.1 Heuristic DR (existing implementation)

The shipping implementation in **ringbrush\_coverage/core.py** treats the per-window mean of the in-plane accelerometer as a gravity proxy and feeds the residual through a damped integrator,

with two small attitude terms that nudge the position by yaw and pitch differences. Concretely, for each window of samples  $\{(\mathbf{a}_i, \phi_i, \theta_i, \psi_i)\}_{i=0}^{N-1}$ :

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**Algorithm 1** Heuristic dead reckoning (per window).

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1:  $\bar{a}_x \leftarrow \text{mean}\{a_{x,i}\}, \quad \bar{a}_y \leftarrow \text{mean}\{a_{y,i}\}$ 
2:  $v_x, v_y, p_x, p_y \leftarrow 0$ 
3: for  $i = 1, \dots, N - 1$  do
4:    $\delta a_x \leftarrow a_{x,i} - \bar{a}_x; \quad \delta a_y \leftarrow a_{y,i} - \bar{a}_y$ 
5:    $v_x \leftarrow 0.92 \cdot v_x + 2.00 \cdot \delta a_x \cdot \Delta t_i$ 
6:    $v_y \leftarrow 0.92 \cdot v_y - 2.00 \cdot \delta a_y \cdot \Delta t_i$ 
7:    $p_x \leftarrow p_x + v_x \cdot \Delta t_i + 0.0015 \cdot \text{circDelta}(\psi_i, \psi_{i-1})$ 
8:    $p_y \leftarrow p_y + v_y \cdot \Delta t_i$ 
9: end for
10: return  $(p_x, p_y)$ 

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The four constants (0.92 damping, 2.00 acceleration scale, 0.0015 yaw contribution, and 0 pitch contribution) were grid-searched against three labelled motion sessions (up-down, left-right, inside-outside) so that the P90 of  $|p_x|$  and  $|p_y|$  matched design targets in the cursor’s normalised visualisation space  $([0, 1] \times [0, 1])$ . The output is therefore unitless; it lives directly in viz coordinates.

## 2.2 AEOLUS DR (faithful port of Radeta 2023)

AEOLUS performs gravity removal via the device orientation, then integrates twice with a ZVU step, then applies a heading projection. The port in `core.py` skips the Madgwick fusion stage because the ring already emits Euler angles directly. We assume the device’s roll and pitch follow the ZYX intrinsic Tait-Bryan convention with world  $+z$  up; under that convention, the expected gravity vector in body frame is

$$\mathbf{g}_b(\phi, \theta) = g \begin{pmatrix} -\sin \theta \\ \cos \theta \sin \phi \\ \cos \theta \cos \phi \end{pmatrix}, \quad g = 9.81 \text{ m/s}^2. \quad (1)$$

The linear acceleration is  $\mathbf{a}_{\text{lin},i} = \mathbf{a}_i - \mathbf{g}_b(\phi_i, \theta_i)$ .

The drift reduction (Algorithm 2) is transcribed *verbatim* from [1, Algorithm 1]: when the linear acceleration along an axis is near zero the velocity is decayed, otherwise the velocity is *replaced* (not accumulated) with  $a \cdot \Delta t$ . This replacement reading is almost certainly a typesetting error in the paper, but we follow the letter rather than the spirit of the text here. We return to this point in Section 6.

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**Algorithm 2** AEOLUS drift reduction [1, Algorithm 1], applied per axis.

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1: if  $|a| < \varepsilon$  then  $\triangleright \varepsilon = 0.05 \text{ m/s}^2$ 
2:    $v \leftarrow v \cdot \beta$   $\triangleright \beta = 0.55 \text{ for } x, y; 0.80 \text{ for } z$ 
3: else
4:    $v \leftarrow a \cdot \Delta t$   $\triangleright \text{verbatim from the paper}$ 
5: end if

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The position update follows equation 9 of the paper, projecting the body-frame step through the heading angle  $\psi_i$ :

$$\begin{aligned} p_{x,i} &= \Delta x_i \cdot \cos \psi_i + p_{x,i-1}, & \Delta x_i &= v_x \cdot \Delta t_i, \quad \Delta y_i = v_y \cdot \Delta t_i. \\ p_{y,i} &= \Delta y_i \cdot \sin \psi_i + p_{y,i-1}, \end{aligned} \quad (2)$$

The output is in metres. For the cursor placement inside `analyze_session`, we rescale per session so the P90 of  $|\Delta x|$  and  $|\Delta y|$  matches the heuristic’s design target ( $\approx 0.35$  viz units). The comparison tool plots the raw, un-rescaled metric trajectory.

### 3 Implementation in the codebase

The faithful port is in `ringbrush_coverage/core.py` as the function `estimate_dead_reckoning_aeolus`, alongside session-wide helpers `trajectory_heuristic` and `trajectory_aeolus` that emit one point per sensor reading. A new CLI flag `-dr-method {heuristic,aeolus}` selects the method end-to-end. The session-wide trajectory generators back a dedicated comparison driver at `tools/compare_dead_` which writes a JSON stats summary, a static PNG, and an animated MP4 per input log. The new functions are exercised by five unit tests in `tests/test_core.py` (stationary input, empty input, trajectory length consistency, body-x response, and `analyze_session(dr_method="aeolus")` integration), all passing together with the four pre-existing tests.

### 4 Experimental setup

We ran the comparison driver on four sessions:

- `left-and-right-and-...` (10s, labelled lateral motion).
- `up-and-down-and-...` (29s, labelled vertical motion).
- `inside-and-outside-and-...` (35s, labelled in-out motion).
- `full-session` (62s, freeform full brushing).

For each session, both methods run once on the full sample stream (no windowing, no per-window resets) and accumulate a continuous 2D trajectory. Per-method summary statistics include the cumulative path length, the end-to-start displacement, and the bounding-box diagonal of the path.

### 5 Results

Table 1 reports the per-session stats. Path length and bounding box are reported in each method’s native unit (the heuristic in normalised viz units, AEOLUS in metres); the path/end ratio expresses how much of a method’s travel *cancels* when integrated — low values mean lots of oscillation back to the origin, high values mean a drift-dominated path.

Table 1: Per-session DR statistics. “path/end” is end-displacement divided by total path length; lower is more oscillatory, higher is more drift-dominated.

| Session        | Heuristic (viz units) |       |          | AEOLUS (metres) |      |          |
|----------------|-----------------------|-------|----------|-----------------|------|----------|
|                | path                  | bbox  | end/path | path            | bbox | end/path |
| left-right     | 23.7                  | 3.82  | 0.009    | 1.63            | 1.21 | 0.73     |
| up-down        | 30.2                  | 2.97  | 0.007    | 1.55            | 1.34 | 0.86     |
| inside-outside | 14.7                  | 2.77  | 0.007    | 3.35            | 3.33 | 0.99     |
| full-session   | 109.7                 | 42.79 | 0.015    | 7.11            | 3.73 | 0.47     |

The qualitative shape of each session is shown in Figures 1–4. Each panel uses independent, data-fit axes, so the shapes are directly comparable but the magnitudes are not.

Dead reckoning comparison — 2026-04-20\_0958\_left-and-right-and-left-and-right-and

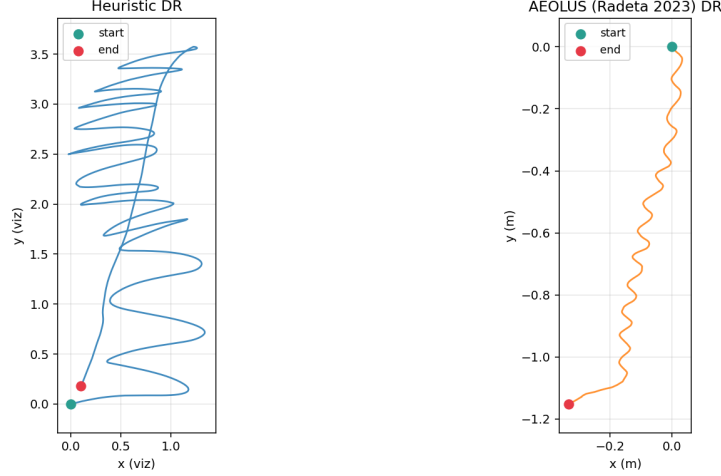


Figure 1: Lateral brushing session. The heuristic (left) cleanly captures the left-right oscillation; AEOLUS (right) drifts roughly south-southwest with small ripples on top.

Dead reckoning comparison — 2026-04-12\_2127\_up-and-down-and-up-and-down-and

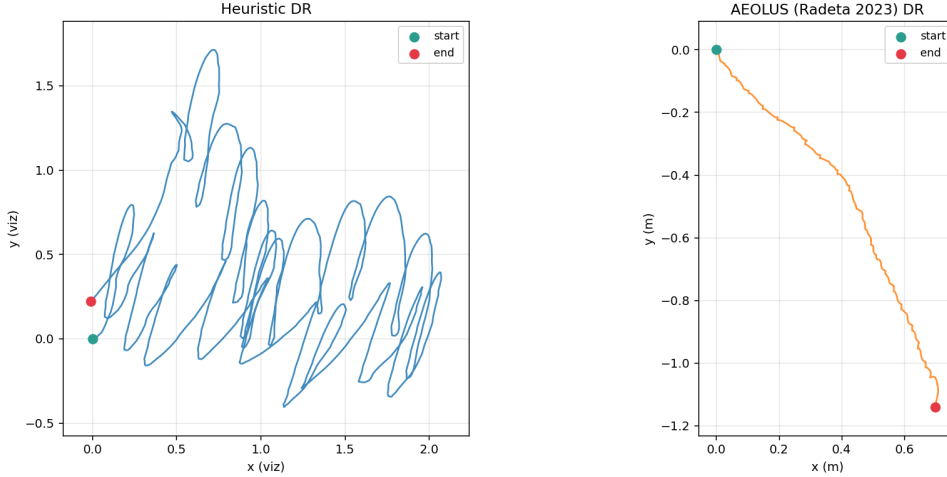


Figure 2: Vertical brushing session. The heuristic produces a recognisable “saw-tooth” fan of strokes; AEOLUS reduces this to a smooth drift toward the lower right.

## 6 Discussion

**Why AEOLUS drifts so heavily on simple motion clips.** The ZVU step in Algorithm 2 only fires when  $|a_{\text{lin}}| < 0.05 \text{ m/s}^2$ , i.e. essentially never during active brushing: the ring is in constant low-grade motion. Without the ZVU intervening, the literal “replacement” velocity assignment  $v \leftarrow a \cdot \Delta t$  produces a small but non-zero step whose sign is set by the *current* linear acceleration. Because brushing motion is roughly periodic but not perfectly symmetric, even a small DC bias in  $\mathbf{a}_{\text{lin}}$  accumulates monotonically over the session. This is exactly the path-versus-end signature we see in the left/right and up/down rows of Table 1: AEOLUS’ end-displacement is 73%–99% of its path length, meaning almost everything that’s integrated stays integrated.

**Why the heuristic looks “right” on those same clips.** The heuristic resets state at the window boundary and subtracts the per-window mean accelerometer reading. Because the brushing motion is oscillatory within a window, this mean is a good proxy for the gravity projection and any drift-inducing DC bias, so the integrator sees a zero-mean signal and produces a zero-mean

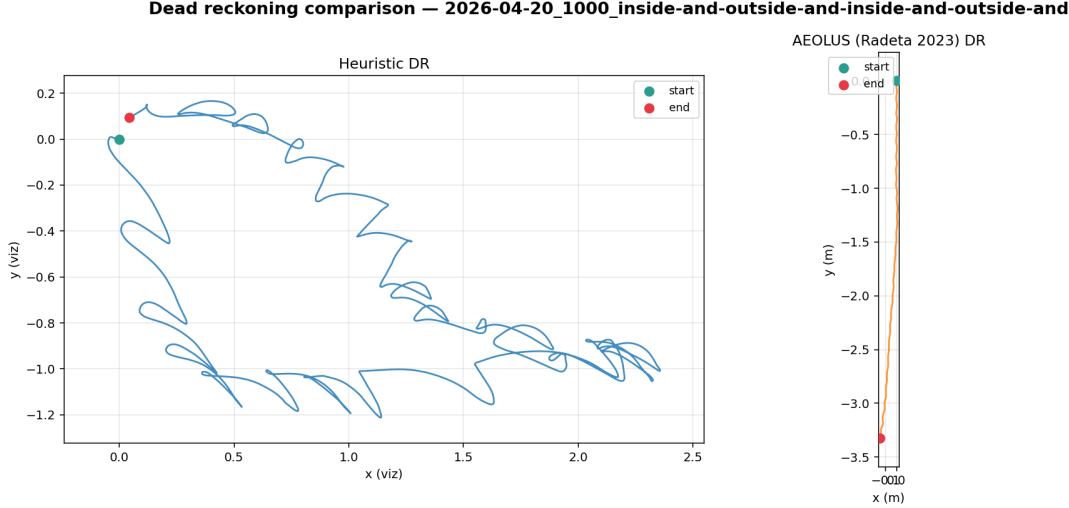


Figure 3: Inside-outside brushing session. The heuristic shows a curving sweep back to roughly the origin; AEOLUS collapses to an almost-1D drift along  $y$  (note the AEOLUS  $x$ -range is so small that the panel is nearly empty).

trajectory. This is what gives the heuristic its 0.7–1.5% end/path ratios across all single-motion sessions: each window’s contribution is bounded and centred. The cost is non-metric output: the path length 109.7 “viz units” on the full session is not in metres or any other physical unit, and the four constants in Algorithm 1 are tuned for the brushing motion specifically.

**Convention assumption.** The Earth-frame gravity removal in Equation 1 assumes the ring reports Euler angles in the ZYX intrinsic Tait-Bryan convention. If the device uses a different rotation order or sign convention, gravity is not fully removed and a residual constant acceleration appears in  $\mathbf{a}_{\text{lin}}$ . Sanity check from the idle calibration: with mean roll  $\approx -134.1^\circ$  and mean pitch  $\approx 31.4^\circ$ , the expected  $\mathbf{g}_b$  under ZYX is approximately  $(-5.11, -6.02, -5.83)$ , while the measured idle mean is  $(5.09, -2.87, 7.82)$ . The magnitudes are close (both  $\approx 9.8 \text{ m/s}^2$ ) but the directions do not match: the device’s convention is not vanilla ZYX. Tuning candidates if the residual matters for downstream use: try YXZ/ZXY orderings, or flip the sign of roll and pitch before constructing the gravity vector.

**Faithfulness versus realism.** We followed the printed Algorithm 1 of [1] literally: the else branch is a replacement,  $v \leftarrow a \cdot \Delta t$ , not an accumulation  $v \leftarrow v + a \cdot \Delta t$ . The replacement form has the effect of high-passing the velocity signal: it forgets prior acceleration entirely the moment motion resumes, which dampens the explosive drift that the double-integration form would produce but also throws away the integrating behaviour that is the whole point of dead reckoning. We believe this is a typo in the paper; under the accumulation reading the AEOLUS paths would be larger and more drift (the paper’s own results indicate 5% displacement error over tens of metres of travel, which is consistent with accumulation plus working ZVU). Swapping to accumulation is a one-line change in `core.py` for users who want to A/B both readings.

**Equation 9 as a 2D oddity.** Equation 9 of [1] reads  $x_i = \Delta x \cos \theta + x_{i-1}$  and  $y_i = \Delta y \sin \theta + y_{i-1}$ . This is not the usual heading-rotation formula (which would mix  $\Delta x$  and  $\Delta y$  through both cos and sin). At  $\theta = 0$  this becomes  $x_i = \Delta x + x_{i-1}$ ,  $y_i = y_{i-1}$ , dropping the  $\Delta y$  contribution entirely. We implement equation 9 as printed; a reader who treats  $\Delta x = \Delta y = \Delta$  (the scalar magnitude of the step) recovers the standard PDR formula. Both interpretations are easy to swap at one line in the `estimate_dead_reckoning_aeolus` function.

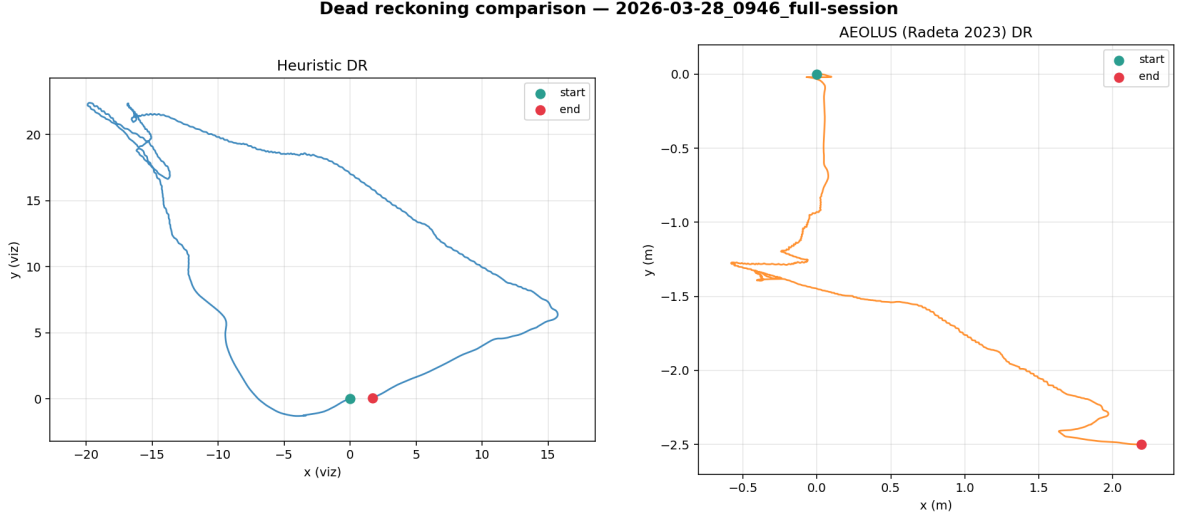


Figure 4: Full freeform brushing session. Here both methods produce a path with non-trivial topology: a wide tour through the upper half (heuristic) and a meander through the right half (AEOLUS), both returning toward the origin.

## 7 Conclusions

For the smart-ring brushing visualisation, the in-house heuristic is the right tool: it captures repetitive in-mouth motion cleanly, its output already lives in viz coordinates, and its four free parameters are explicitly fit to the cursor-placement objective. The AEOLUS port has value mainly as a sanity check and as a starting point for a metric trajectory estimate. With the faithful Algorithm 1 transcription, AEOLUS systematically under-estimates oscillation amplitude. Restoring the standard accumulation form, sharpening the ZVU threshold for ring-specific motion, and verifying the Euler-angle convention would all push the AEOLUS path closer to a metric ground truth, at which point the comparison shifts from “which one looks more reasonable” to “which one predicts brushed area more accurately.” Both directions are easy follow-ups from the code already in place.

## References

- [1] Marko Radeta, Claudio Rodrigues, Francisco Silva, Pedro Abreu, João Pestana, Ngoc Thi Nguyen, Agustin Zuniga, Huber Flores, Petteri Nurmi. Lost in the Deep? Performance Evaluation of Dead Reckoning Techniques in Underwater Environments. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(2), Article 73, June 2023. [doi:10.1145/3596245](https://doi.org/10.1145/3596245).